

Math 2650 – Quiz 03 Concept Practice

Choose the section matching the skill you want to practice. Answers are on the back.

A. Motion Along a Space Curve

An object moves along the curve

$$\mathbf{r}(t) = \langle t^3 + 2t, 4\sqrt{t+1}, 2\ln(t+1) \rangle.$$

- (a) At $t = 0$ the object's engine shuts off, after which it travels with constant velocity $\mathbf{r}'(0)$. Find its position two units of time after the engine shuts off.
- (b) Find the angle between the object's velocity and acceleration vectors at $t = 0$.
- (c) Find an equation of the plane through $\mathbf{r}(0)$ that is perpendicular to the curve at $t = 0$.

B. Tangent Lines and Intersecting Curves

Consider the curve

$$\mathbf{r}(t) = \langle 2t + 1, t^2 - 3, -t + 4 \rangle.$$

- (a) Find all values of t for which the tangent line to $\mathbf{r}(t)$ passes through the point

$$Q = (5, 0, 2).$$

- (b) A second curve is

$$\mathbf{s}(u) = \langle 3 + u, -2 + 2u + u^2, 3 + 2u - u^2 \rangle.$$

The curves intersect at $\mathbf{r}(1) = \mathbf{s}(0)$. Let θ be the angle between the tangent vectors to the two curves at their point of intersection. Find $\cos \theta$.

C. Quadric Surfaces and Cross Sections

Consider the surface

$$4x^2 + 9y^2 - 8x + 36y - 12z + 4 = 0.$$

- (a) Rewrite the equation in standard form. Identify the quadric surface, state its axis of symmetry, and give the direction in which it opens.
- (b) Identify the conic section obtained by intersecting the surface with each of the planes

$$x = 1, \quad y = -2, \quad z = 0.$$

- (c) Find the area of the cross section obtained when $z = 0$.

D. Spherical and Cartesian Descriptions of a Surface

A surface is given in spherical coordinates by

$$\rho = \frac{8 \sin \phi \sin \theta}{1 + 3 \cos^2 \phi}.$$

- (a) Rewrite the equation in Cartesian coordinates.
- (b) Put the Cartesian equation into standard form and identify the quadric surface.
- (c) Give its center and semiaxis lengths.

Answer Key

Final answers only

A. Motion Along a Space Curve

(a)

$$(4, 8, 4)$$

(b)

$$\theta = \arccos\left(-\frac{\sqrt{15}}{5}\right)$$

(c)

$$x + y + z = 4$$

B. Tangent Lines and Intersecting Curves

(a)

$$t = 1, 3$$

(b)

$$\cos \theta = \frac{4}{9}$$

C. Quadric Surfaces and Cross Sections

(a)

$$\frac{(x-1)^2}{9} + \frac{(y+2)^2}{4} - \frac{z}{3} = 1$$

Elliptic paraboloid

Axis of symmetry:

$$x = 1, \quad y = -2$$

which is parallel to the z -axis, and the surface opens in the

positive z -direction.

(b)

$$x = 1 : \text{parabola}, \quad y = -2 : \text{parabola}, \quad z = 0 : \text{ellipse}$$

(c)

$$6\pi$$

D. Spherical and Cartesian Descriptions of a Surface

(a)

$$x^2 + y^2 + 4z^2 = 8y$$

(b)

$$\frac{x^2}{16} + \frac{(y-4)^2}{16} + \frac{z^2}{4} = 1$$

Ellipsoid

(c) Center:

$$(0, 4, 0)$$

with semiaxis lengths

$$4, 4, 2.$$